E-COMMERCE SALES ANALYSIS

MYSQL

TOTAL REVENUE

```sql
SELECT ROUND(SUM(payment_value), 2) AS total_revenue
FROM olist_order_payments_dataset;
```

total_revenue
1674079.03

MONTHLY REVENUE TREND

```sql
SELECT
    YEAR(o.order_purchase_timestamp)  AS year,
    MONTH(o.order_purchase_timestamp) AS month,
    ROUND(SUM(p.payment_value), 2)    AS monthly_revenue
FROM olist_orders_dataset o
JOIN olist_order_payments_dataset p ON o.order_id = p.order_id
GROUP BY year, month
ORDER BY year, month;
```

year	month	monthly_revenue
2016	10	4309.62
2017	1	6878.92
2017	2	26146.53
2017	3	25218.09
2017	4	24431.68
2017	5	35920.35
2017	6	26210.29
2017	7	29480.69
2017	8	44950.08
2017	9	44513.04
2017	10	43071.14
2017	11	63035.24
2017	12	51210.86
2018	1	60966.16
2018	2	71967.2

TOTAL ORDERS BY STATUS

```sql
SELECT
    order_status,
    COUNT(*) AS total_orders
FROM olist_orders_dataset
GROUP BY order_status
ORDER BY total_orders DESC;
```

order_status	total_orders
delivered	96478
shipped	1107
canceled	625
unavailable	609
invoiced	314
processing	301
created	5
approved	2

AVERAGE DELIVERY TIME (IN DAYS)

```sql
SELECT
    ROUND(AVG(DATEDIFF(order_delivered_customer_date,
                       order_purchase_timestamp)), 1) AS avg_delivery_days
FROM olist_orders_dataset
WHERE order_status = 'delivered'
  AND order_delivered_customer_date IS NOT NULL;
```

avg_delivery_days
12.5

LATE VS ON-TIME DELIVERIES — IMPACT ON REVIEW SCORE

```sql
SELECT
    CASE
        WHEN order_delivered_customer_date > order_estimated_delivery_date
            THEN 'Late'
        ELSE 'On Time or Early'
    END AS delivery_status,
    ROUND(AVG(r.review_score), 2) AS avg_review_score,
    COUNT(*)                      AS total_orders
FROM olist_orders_dataset o
JOIN olist_order_reviews_dataset r ON o.order_id = r.order_id
WHERE o.order_delivered_customer_date IS NOT NULL
  AND o.order_estimated_delivery_date IS NOT NULL
GROUP BY delivery_status;
```

delivery_status	avg_review_score	total_orders
On Time or Early	4.16	255
Late	2.50	24

TOP 10 SELLERS BY REVENUE

```sql
SELECT
    oi.seller_id,
    s.seller_city,
    s.seller_state,
    COUNT(DISTINCT oi.order_id) AS total_orders,
    ROUND(SUM(oi.price), 2)     AS total_revenue
FROM olist_order_items_dataset oi
JOIN olist_sellers_dataset s ON oi.seller_id = s.seller_id
GROUP BY oi.seller_id, s.seller_city, s.seller_state
ORDER BY total_revenue DESC
LIMIT 10;
```

seller_id	seller_city	seller_state	total_orders	total_revenue
53243585a1d6dc2643021fd1853d8905	lauro de freitas	BA	60	33434.76
4869f7a5dfa277a7dca6462dcf3b52b2	guariba	SP	140	27610.84
fa1c13f2614d7b5c4749cbc52fecda94	sumare	SP	74	23132.92
4a3ca9315b744ce9f8e9374361493884	ibitinga	SP	203	22693.4
7c67e1448b00f6e969d365cea6b010ab	itaquaquecetuba	SP	157	22643.46
1025f0e2d44d7041d6cf58b6550e0bfa	sao paulo	SP	177	20887.09
7e93a43ef30c4f03f38b393420bc753a	barueri	SP	45	19256.3
da8622b14eb17ae2831f4ac5b9dab84a	piracicaba	SP	174	18445.08
7a67c85e85bb2ce8582c35f2203ad736	sao paulo	SP	132	17336.55
955fee9216a65b617aa5c0531780ce60	sao paulo	SP	164	15047.91

MOST POPULAR PAYMENT METHOD

```sql
SELECT
    payment_type,
    COUNT(*)                        AS total_transactions,
    ROUND(SUM(payment_value), 2) AS total_value
FROM olist_order_payments_dataset
GROUP BY payment_type
ORDER BY total_transactions DESC;
```

payment_type	total_transactions	total_value
6d33559681ad82d2e835b1cb587d2ce3	2	68.44
-46.59035813849616	2	269.18
71540d636652fc682032d3b3c0a180c1	1	64.09
-46.52727873105775	1	96.71
-46.53013786011885	1	8.58
-46.52928120996594	1	107.1
9007d9a8a0d332c61d9dd611fa341f4b	1	40.76
3af242f6518c23ebab15619fdc356e43	1	44.18
4daac62d42cfbad476c3e42791411412	1	44.29
-46.542844927837734	1	119.8
-46.505467534557326	1	243.45
-46.49776233870935	1	75.65
a88d7a0d11f1f620053197f94584d6d2	1	506.17
-46.512950809740886	1	36.3
-46.65950430030583	1	47.2

REVENUE BY STATE

```sql
SELECT
    c.customer_state,
    ROUND(SUM(p.payment_value), 2) AS total_revenue
FROM olist_orders_dataset o
JOIN olist_customers_dataset      c ON o.customer_id = c.customer_id
JOIN olist_order_payments_dataset p ON o.order_id    = p.order_id
GROUP BY c.customer_state
ORDER BY total_revenue DESC;
```

customer_state	total_revenue
SP	373197.22
RJ	117377.39
MG	111767.19
RS	58362.14
PR	52386.12
BA	34350.07
SC	32418.5
ES	26026.03
GO	22313.81
CE	20845.64
DF	20799.47
PA	15803.2
PE	14749.48
PB	9797.06
MS	8518.15

REVIEW SCORE DISTRIBUTION

```sql
SELECT
    review_score,
    COUNT(*) AS total_reviews
FROM olist_order_reviews_dataset
GROUP BY review_score
ORDER BY review_score DESC;
```

review_score	total_reviews
5	164
4	39
3	30
2	8
1	38

REPEAT CUSTOMERS

```sql
SELECT
    COUNT(*) AS repeat_customers
FROM (
    SELECT
        c.customer_unique_id,
        COUNT(o.order_id) AS order_count
    FROM olist_customers_dataset c
    JOIN olist_orders_dataset o ON c.customer_id = o.customer_id
    GROUP BY c.customer_unique_id
    HAVING order_count > 1
) AS repeat_buyers;
```

repeat_customers
2997

TOP 10 CUSTOMERS BY TOTAL SPENDING

```sql
SELECT
    c.customer_unique_id,
    c.customer_city,
    c.customer_state,
    COUNT(DISTINCT o.order_id)     AS total_orders,
    ROUND(SUM(p.payment_value), 2) AS total_spent
FROM olist_customers_dataset c
JOIN olist_orders_dataset         o ON c.customer_id = o.customer_id
JOIN olist_order_payments_dataset p ON o.order_id    = p.order_id
GROUP BY c.customer_unique_id, c.customer_city, c.customer_state
ORDER BY total_spent DESC
LIMIT 10;
```

customer_unique_id	customer_city	customer_state	total_orders	total_spent
5e713be0853d8986528d7869a0811d2b	belem	PA	1	8085.48
a826e173df23e6620170be7da0233ef5	belo horizonte	MG	1	4667.94
6cba2fea42066c8781a30cd92826317c	contagem	MG	1	4121.02
9654d6df6a63f5a9b61dd5ba0e0d8286	sobral	CE	1	3783.26
d91f92777fc8e939f395a1f6e5f33ebb	porto velho	RO	1	3656.82
3ac427a63b9d1586ffa6f85c2e924717	sao paulo	SP	1	3358.24
6411590d91c48640cb07e72fbb4a359e	rio verde	GO	1	3239.46
8a7dbdbf6eeb386835be7ee745661739	rio de janeiro	RJ	1	3210.28
5eac32fd7c8fbf52e46c891228ac3067	santo andre	SP	1	2827
bd20b118b1f383a4978c8e6e326ccbd4	guarapuava	PR	1	2787.87

THANK YOU